

Prakriti Shakya

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Technical Skills

Languages: Python, Java, SQL, Oracle PL/SQL, JavaScript, C++, Bash

Data Engineering: Spark, PySpark, Kafka, Airflow, dbt, ETL/ELT Pipelines, Streaming Pipelines, Data Modeling, Snowflake, BigQuery, Distributed Systems, Real-Time Data Processing

AI & Machine Learning: OpenAI API, Generative AI, LLM Applications, Prompt Engineering, RAG, NLP, AI Chatbots, Embedding Pipelines, Semantic Search, Predictive Analytics, Model Deployment

Cloud & DevOps: AWS (Lambda, Bedrock, Glue, SageMaker), GCP (BigQuery, GCS, Dataproc), Docker, Kubernetes, CI/CD, Cloud-Native Systems, Linux, Bash Scripting, Infrastructure Automation

Backend & APIs: FastAPI, REST APIs, Backend Development, Microservices, Event-Driven Architectures, API Integration

Security & Workflow: Data Governance, Schema Validation, Cloud Security, Agile, Git, Jira, Cross-functional Collaboration

Experience

Computer Science Tutor – California State University, Northridge Jul 2025 – Present

- Teach Data Structures and Algorithms including recursion, trees, graphs, dynamic programming, complexity analysis, and debugging techniques.
- Mentor in performance optimization, maintainable coding practices and distributed systems concepts.

Associate Data Engineer – Tata Consultancy Services Oct 2021 – Jul 2024

- Maintained Spark-based ETL/ELT pipelines processing 500M+ records in distributed cloud production environments.
- Built fault-tolerant streaming and batch workflows using Spark, Kafka, and Airflow, reducing runtime by nearly 20%.
- Developed Airflow DAGs with SLA alerting, retry logic, automated monitoring, observability workflows, and production incident resolution.
- Implemented schema validation, automated data quality checks, governance workflows, and monitoring systems to improve reliability and integrity.
- Collaborated with analysts, QA teams, and engineers to deliver scalable analytical data workflows.

Projects

Intelligent Data Pipeline + AI Insights System Mar 2026

- Designed an end-to-end AI-powered data platform using Kafka, Spark, Airflow, and cloud-native architectures for real-time analytics workflows.
- Built scalable ETL and event-driven streaming pipelines using BigQuery and Snowflake for downstream analytics.
- Developed ML workflows for predictive analytics, anomaly detection on large-scale streaming datasets.
- Integrated OpenAI/Llama-based LLM workflows using embedding pipelines, prompt engineering, semantic search, and RAG techniques for intelligent querying.
- Developed FastAPI microservices and Streamlit/React dashboards for real-time monitoring, AI-generated insights, and analytics visualization.
- Deployed cloud-native workloads on AWS using Docker, CI/CD pipelines, automated monitoring, and automation workflows.

Event Analytics Pipeline – Spark + Kafka + dbt Jan 2026

- Built scalable real-time event processing pipelines using Spark, Kafka, and Airflow for high-volume streaming workloads.
- Developed dbt transformation models with reusable macros, incremental materializations, and Snowflake-based analytical data modeling.
- Modeled consumption-ready datasets for downstream reporting, BI dashboards, AI workflows, and data visualization.
- Implemented SLA alerting, automated retries, anomaly detection, and data quality monitoring to improve reliability and scalability.

Publications

- **Evaluating the Vulnerability of Deepfake Image Detection Models to Adversarial Manipulations**, IEEE ISDFS 2026
- Evaluated deep learning models under adversarial attacks and PGD-based perturbations to analyze robustness and generalization performance.

Education

California State University, Northridge

M.S. Computer Science

Aug 2024 – May 2026

GPA: 3.9/4.0 (Graduated with Distinction)